

PhysiLearn Engine

Investment Pitch Memorandum

Next-Generation AI Adaptive Learning for STEM

The Problem: Static & Unpersonalized Education

Traditional education lacks personalization, leaving students struggling with complex, dynamic concepts. In fields like physics, static textbook problems fail to effectively teach intuitive understanding of systems in motion. Without interactive, adaptive feedback, dropout rates in foundational STEM courses remain unacceptably high.

Our Solution: Core Tech for EdTech

PhysiLearn is developing a proprietary **AI-driven physics game engine** specifically designed for the EdTech sector. We license our core technology to educational platforms (B2B2C model), allowing them to instantly generate personalized, interactive simulation environments.

Instead of reading about **classical mechanics, torque, or oscillatory motion**, students interact with real-time simulations. The AI layer monitors student interactions and adaptively scales the difficulty of the physics scenarios, utilizing advanced prompt engineering methodologies (such as the CLEAR framework) to guide AI tutoring agents that assist the student in real-time.

Technology Advantage

Our competitive moat lies in our highly optimized underlying architecture, built for extreme performance in web and local environments. By avoiding standard, bloated game engines, we deliver a lightweight, specialized solution.

- **Optimized Core Logic:** Written in C++ and Python, ensuring high-speed processing of complex hydrostatic forces and collision physics.
- **Advanced Object Management:** Leveraging custom data structures—specifically tailored Heap structures and AVL trees—to efficiently manage thousands of concurrent game objects and state changes without performance degradation.
- **AI Agent Evaluation:** Continuous evaluation frameworks ensure our NLP tutoring agents provide precise, context-aware assistance rather than generic answers.

C++ & Python

AI / NLP Agents

Adaptive Learning

Heap & AVL Object Management

\$10B+

TARGET EDTECH MARKET

40%

INCREASE IN RETENTION

B2B2C

BUSINESS MODEL

Proposal for NextGen Ventures

We are seeking a **\$300,000 Seed Investment** to scale our engineering team and finalize our B2B APIs for initial EdTech launch partners.

This opportunity perfectly aligns with NextGen Ventures' thesis of investing in highly scalable, core educational technology that solves the lack of personalized learning through AI and Adaptive Learning.